

DRAWING LIST FOR TEST PACKAGE

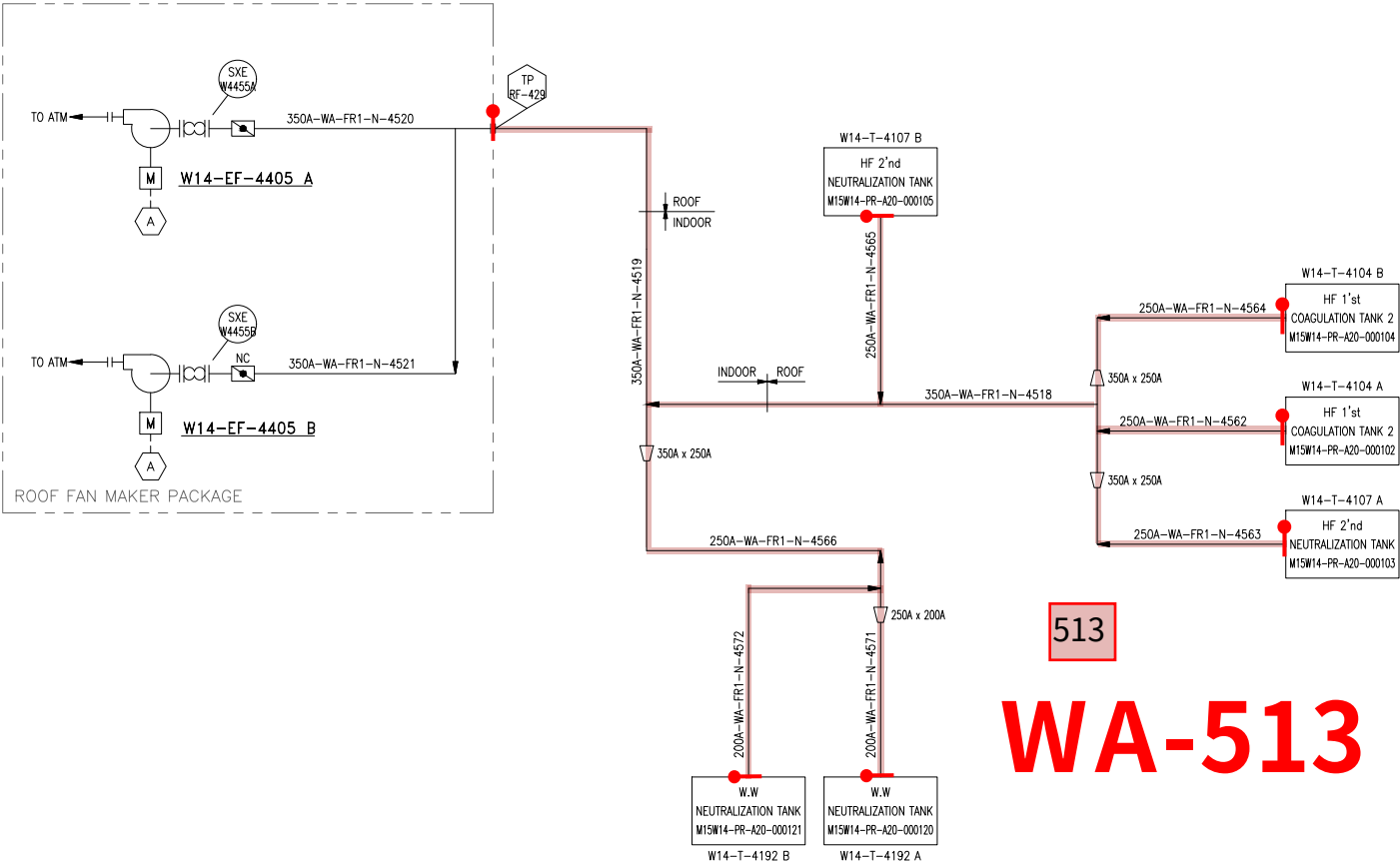
WA-513

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

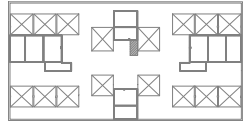
No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	513	3FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
2	513	4FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
3	513	5FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
4	513	5FA	350A-WA-FR1-N-4518-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
5	513	5FA	350A-WA-FR1-N-4519-01	1 OF 1	1	CJ-M15W14-PR-A20-169		
6	513	RFA	250A-WA-FR1-N-4562-01	1 OF 1	0	CJ-M15W14-PR-A20-000102/169		
7	513	RFA	250A-WA-FR1-N-4563-01	1 OF 1	0	CJ-M15W14-PR-A20-000103/169		
8	513	RFA	250A-WA-FR1-N-4564-01	1 OF 1	0	CJ-M15W14-PR-A20-000104/169		
9	513	RFA	250A-WA-FR1-N-4565-01	1 OF 1	0	CJ-M15W14-PR-A20-000105/169		
10	513	RFA	350A-WA-FR1-N-4518-01	1 OF 2	0	CJ-M15W14-PR-A20-169		
11	513	RFA	350A-WA-FR1-N-4518-02	2 OF 2	0	CJ-M15W14-PR-A20-169		
12	513	2FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
13	513	2FA	250A-WA-FR1-N-4571-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
14	513	2FA	250A-WA-FR1-N-4572-01	1 OF 1	0	CJ-M15W14-PR-A20-169		

AI(594x841mm ±2mm)



ROOF FAN MAKER PACKAGE	
ITEM NO.	W14-EF-4405 A/B
SERVICE	ROOF FAN
Q'TY	1 + 1
TYPE	TURBO
CAPA. CMM x mmAq	100 x 300
MAT'L	FRP/ST316
POWER kW	11

대외비



KEY PLAN (RF)

NOTE

1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE

제도	이인학
DRAWN BY	
검토	한진희
CHECKED BY	
승인	박동영
APPROVED BY	

제작사
SK 에코플랜트

공사명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
ROOF FAN

축척
SCALE
NONE

날짜
DATE
2025. 08. 14

PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000169	1A

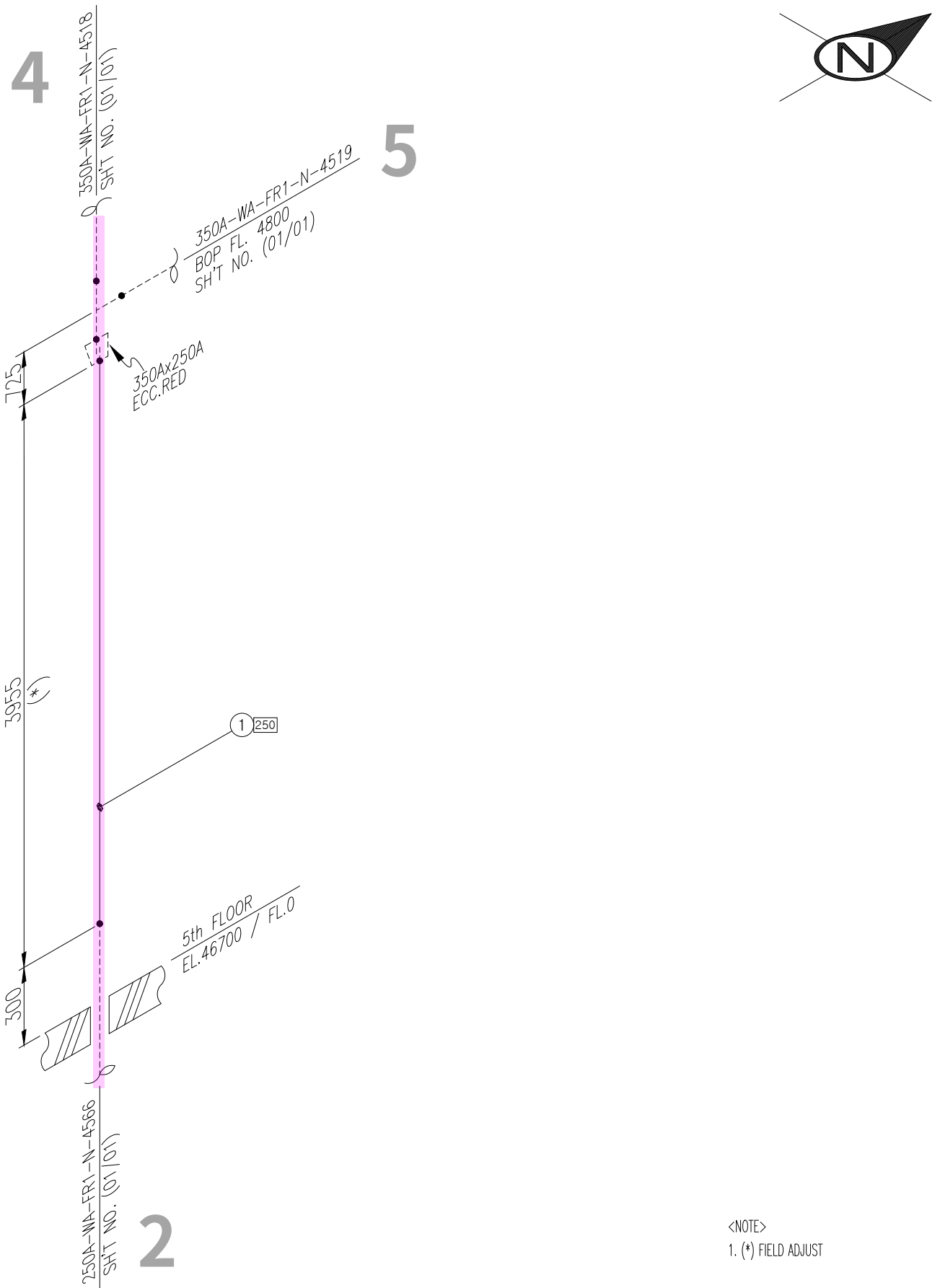
[illegible]

2

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	AMB	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A								WA	22330D	
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000169		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	250A-WA-FR1-N-4566	01 / 01	0

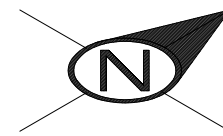
정답 정답은 ②이다.



<NOTE>
1. (*) FIELD ADJUST

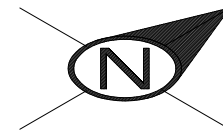
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR SK ecoplant	PROJECT NAME		
	THICK.	—	DESIGN	0.8	60	UNIT	5FA										M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.										COMMODITY	JOB NO.		
						CJ-M15W14-PI-A30-05006-A		△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			WA	22330D		
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			LINE NO.	250A-WA-FR1-N-4566	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000169											01 / 01	0	

[illegible]

CLIENT				4			
							
CONTRACTOR				PROJECT NAME			
				M15X Ph4 Project			
COMMODITY				JOB NO.			
WA				22330D			
LINE NO.				SH'T NO.		REV.	
350A-WA-FR1-N-4518				01 / 01		0	

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1													
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	-	DESIGN	1	AMB	UNIT	5FA						
			OPERATING	ATM	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-050009-C		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동	
						REF. P & ID CJ-M15W14-PR-A20-000169							
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP		



1. (*) FIELD ADJUST

[illegible]

CLIENT

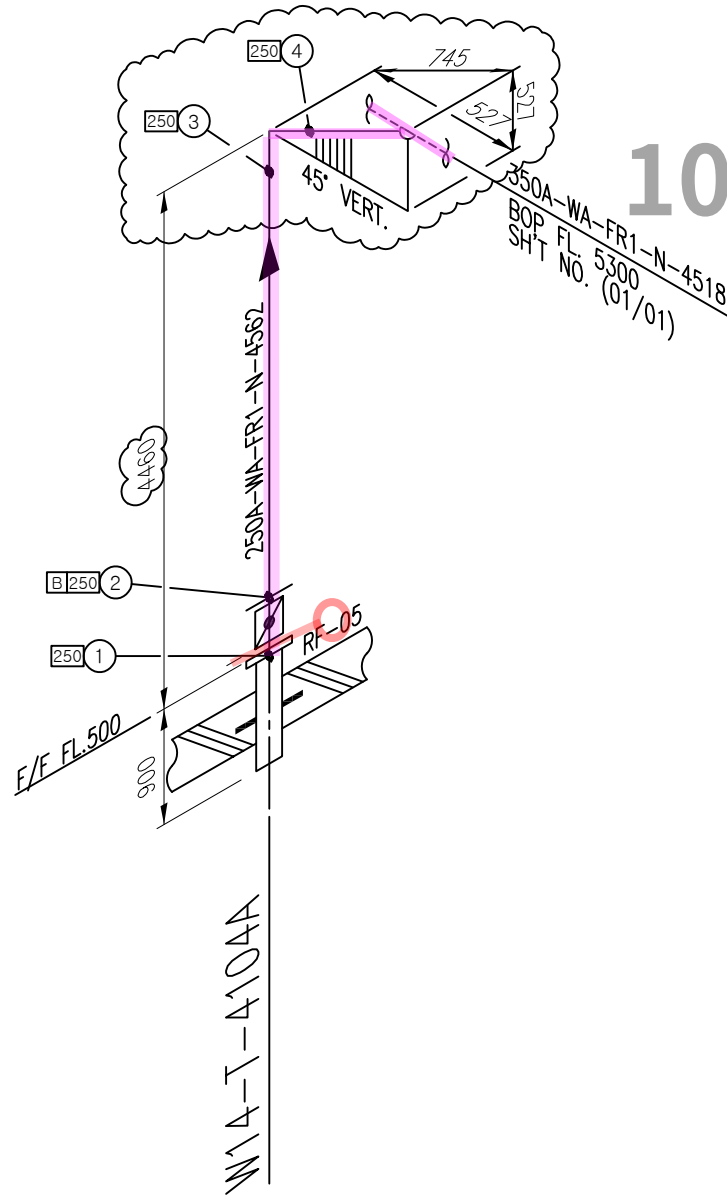


5

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	AMB	UNIT	5FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-050005-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	WA	22330D	
						REF. P & ID CJ-M15W14-PR-A20-000169		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	350A-WA-FR1-N-4519	SH'T NO. 01 / 01

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

[illegible]

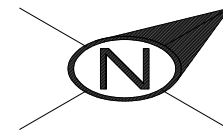
<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2L	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	AMB	UNIT	RFA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-R10008-A							COMMODITY	JOB NO.		
						REF. P & ID	CJ-M15W14-PR-A20-000169		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	250A-WA-FR1-N-4562	SH'T NO.	REV.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			01 / 01	0

정고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



1. (*) FIELD ADJUST

CLIENT  7

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 복제, 제3자에게 공개, 누설 또는 사용함으로써 회사에 손해가 발생할 경우 응징 조치될 수 있으며, 회사는 응징 조치를 취할 권리가 있습니다.

[illegible]

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

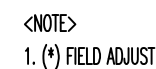


8

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME		
	THICK.	—	DESIGN	1	AMB	UNIT RFA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10009-A							COMMODITY	WA		
						REF. P & ID CJ-M15W14-PR-A20-000169		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	250A-WA-FR1-N-4564	SH'T NO. 01 / 01	REV. 0
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				

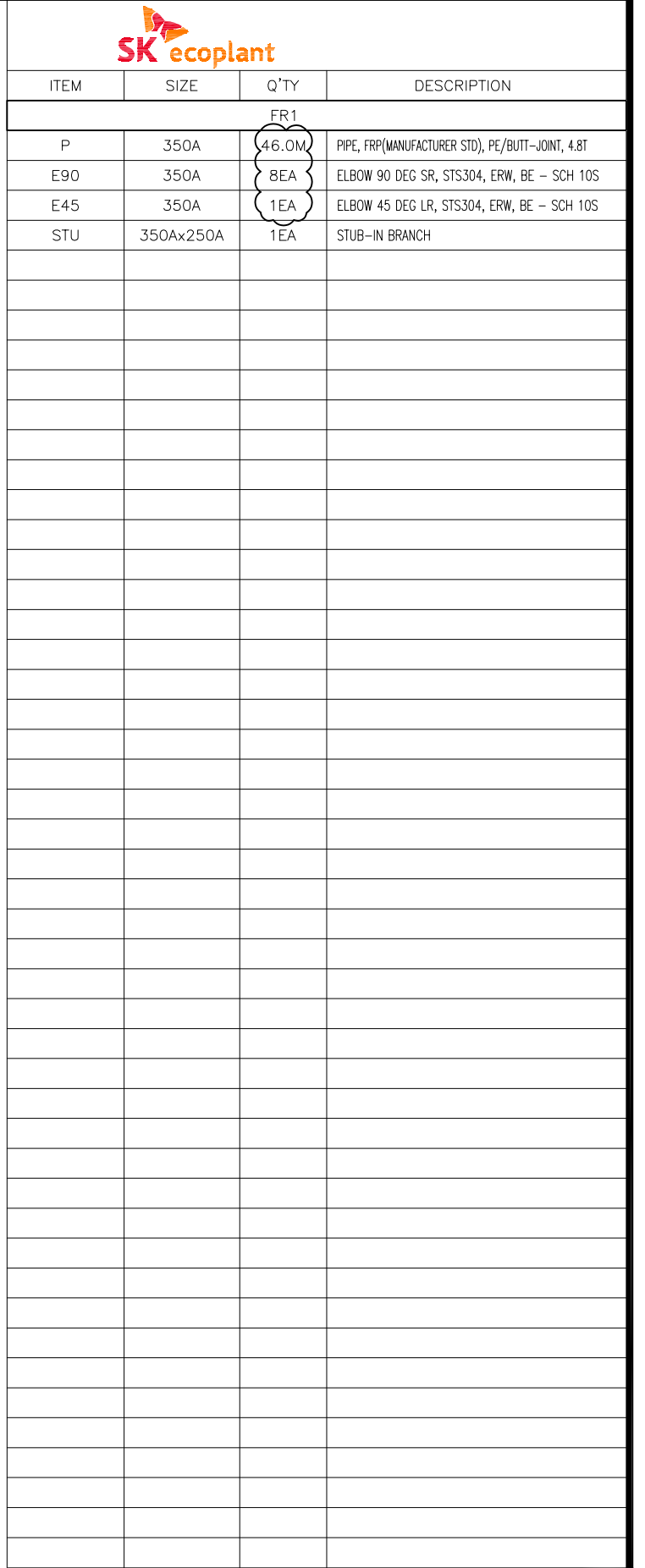


CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WA		22330D	
LINE NO.	350A-WA-FR1-N-4518	SH'T NO. 01 / 02	REV. 0

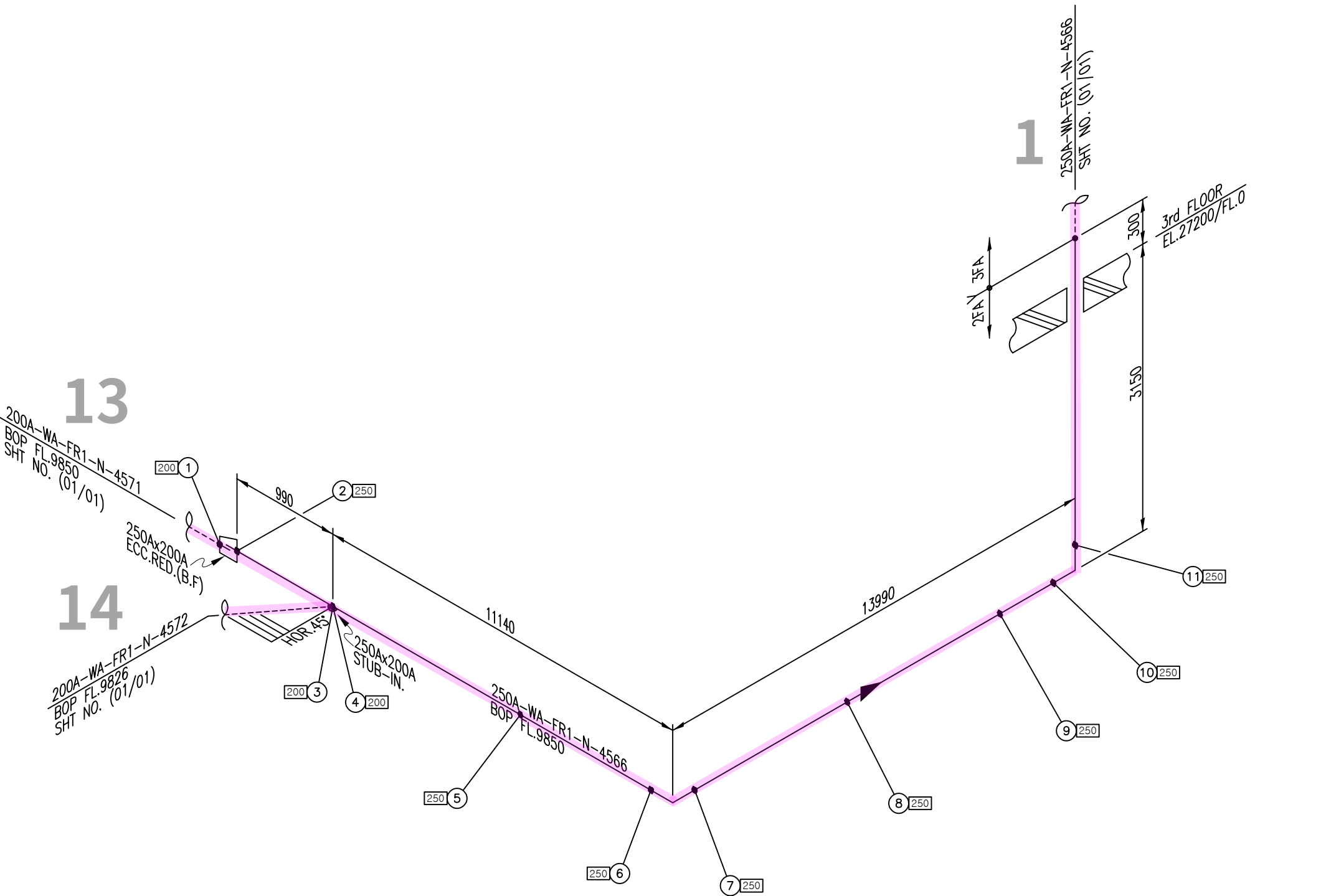


INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	1	AMB	UNIT	RFA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-R10001-B								WA	22330D	
						REF. P & ID								LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000169								350A-WA-FR1-N-4518	01 / 02	0
									25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 제3자의 사용, 복제, 제3자에게 공개, 누설 또는 배포할 수 없으며, 회사는 유보자의 무단 사용으로 피해가 발생했을 경우 유보자에게 민/형사상 책임 추궁할 수 있음



CLIENT		 11	
CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WA		22330D	
LINE NO.	350A-WA-FR1-N-4518	SH'T NO.	REV.
		02 / 02	0



* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR 	PROJECT NAME		
	THICK.	—	DESIGN	1	AMB	UNIT	2FA										M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ—M15W14—PI—A30—020011—E			24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영			COMMODITY	JOB NO.		
						REF. P & ID CJ—M15W14—PR—A20—000169										WA	22330D		
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			LINE NO.	250A—WA—FR1—N—4566	SH'T NO. 01 / 01	REV. 0



ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	200A	24.8M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	200A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
E45	200A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
VP	200A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
F	200A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	200A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS



1. (*) FIELD ADJUST

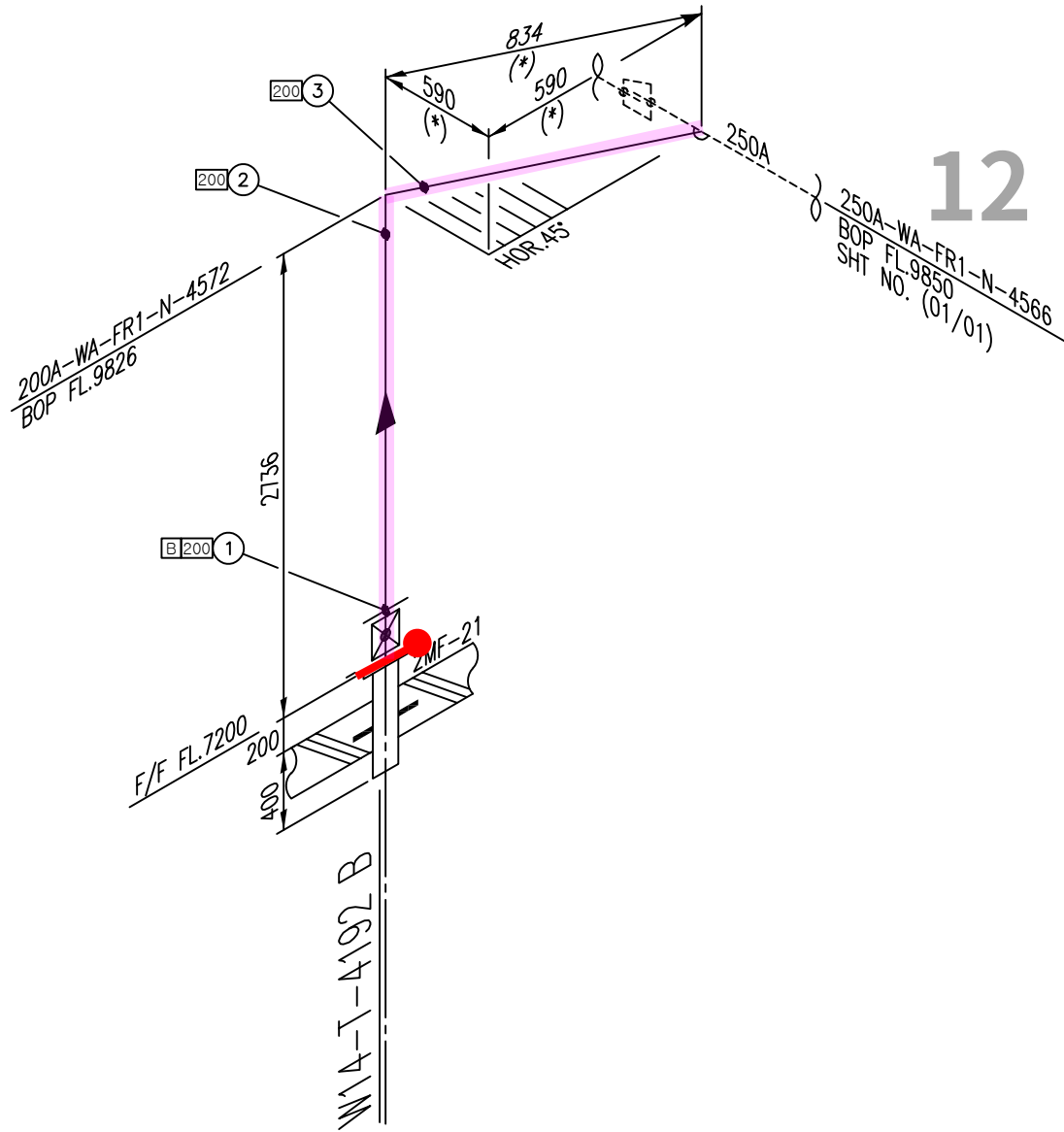
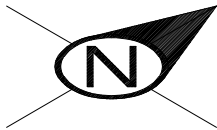
CLIENT



13

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	AMB	UNIT	2FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-020011-E								WA	22330D	
						REF. P & ID			24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000169		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	200A-WA-FR1-N-4571	01 / 01	0

정경호, 최필진, 이원익, SK HYNIX의 기밀비밀 자료를로서, 어떠한 경우에도 동도록에 의외로 상용 목적 제3자에게 공개, 누설, 수출할 수 없으며, 회사내 유·무효자의 불법행위나 사내유출로 피해가 발생할 경우 유·무효자에게 민·형사상 책임이 있음을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR		PROJECT NAME
	THICK.	—	DESIGN	1	AMB	UNIT	2FA									M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020012-E			24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			COMMODITY	WA	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000169										LINE NO.	200A-WA-FR1-N-4572	SH'T NO. 01 / 01